

Abstract

Online voting systems have become increasingly important with the growth of digital services; however, most traditional systems are based on centralized servers and databases. Such systems are often vulnerable to security attacks, data manipulation, and lack of transparency, which can reduce the trust of voters in the election process. Ensuring the integrity and fairness of digital voting therefore remains a major challenge. This project presents a secure and transparent online voting system based on blockchain technology. The proposed system uses Ethereum smart contracts written in Solidity to store and manage voting data in a decentralized and tamper-proof manner. By using blockchain, the system removes the dependency on a central authority and ensures that once a vote is recorded, it cannot be altered or deleted. The system is designed with two main parts: a web-based voting interface for voters and an administrative panel for managing elections and candidates. Voters can securely cast their votes through the web application, while election authorities can create and monitor elections in a transparent way. All voting activities are permanently recorded on the blockchain, which allows results to be verified by anyone. The proposed solution aims to improve the security, transparency, and trustworthiness of online elections. This system can be effectively used in educational institutions, organizations, and small-scale elections where reliable and tamper-free digital voting is required.

[Keywords: Security, Integrity, Tamper-proof, Ethereum, Solidity, Decentralized, Blockchain, Voter, Administrator]

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Abbreviations

API – Application Programming Interface

DApps – Decentralized Applications

DDoS – Denial of Service

DFD – Data Flow Diagram

e-voting – electronic voting

HTTPS – Hypertext Transfer Protocol Secured

OCR – Optical Character Recognition

OTP – One Time Password

QR Code – Quick Response Code

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Chapter 1- Introduction

1.1 Overview of the system

The proposed system is a blockchain-based online voting platform designed to ensure secure, transparent, and tamper-resistant elections. It addresses the limitations of traditional online voting systems, including centralized control, data manipulation risks, and lack of transparency. By integrating blockchain technology with a web-based interface, the system ensures that votes are securely recorded and publicly verifiable without revealing voter identity.

The architecture consists of three primary layers: the user interface layer, the application layer, and the blockchain layer. The user interface is a web-based application that provides access for both voters and administrators. Voters can view active elections and cast votes, while administrators can create elections, manage candidates, and monitor voting activities. The interface is designed for simplicity and accessibility.

The application layer functions as an intermediary between the frontend and the blockchain. It processes user requests, validates inputs, and communicates with smart contracts using blockchain libraries such as ethers.js, ensuring secure and structured interaction with the network.

The blockchain layer forms the core of the system. Smart contracts written in Solidity and deployed on the Ethereum blockchain manage election data, candidate information, and vote records in an immutable manner. Once confirmed, votes cannot be altered or deleted, ensuring integrity and auditability.

Overall, the system provides a decentralized and reliable voting platform suitable for institutional and small-scale elections where transparency, accuracy, and security are essential.

1.2 Problem Statement

The core problem is that traditional online voting systems are centralized, vulnerable to manipulation, and lack transparency, which reduces stakeholder trust in institutional

and organizational elections. There is a need for a decentralized and verifiable solution that can ensure election integrity without depending on a single controlling authority.

1.3 Objectives

1.3.1 Main Objective

The main objective of this project is to design and develop a secure, transparent, and reliable online voting system using blockchain technology. The system aims to ensure that the entire voting process is tamper-proof, publicly verifiable, and free from centralized control, thereby improving trust, security, and fairness in digital elections.

1.3.2 Specific Objective

1. To design and implement a secure blockchain-based voting architecture using Ethereum smart contracts and automated election management.
2. To develop a web-based voting platform with wallet-based authentication that ensures secure voter validation and enforces the one-person-one-vote policy.
3. To integrate OCR-based identity verification and administrative approval mechanisms to validate voter eligibility before participation.
4. To provide a transparent and tamper-proof vote recording and real-time counting mechanism through blockchain technology.
5. To develop an administrative interface that enables election authorities to create elections, manage candidates, and monitor voting activities efficiently.

Chapter 2- Literature Review

2.1 Introduction to Online Voting Systems

Online voting systems have been developed to improve the efficiency, accessibility, and inclusiveness of traditional election processes. Conventional paper-based voting is time-consuming, costly, and susceptible to human error. Electronic voting systems were introduced to automate vote casting and counting while reducing administrative workload [1].

However, voting is a highly sensitive democratic process that requires strong guarantees of integrity, transparency, and reliability. Security and privacy concerns in digital systems have been widely documented, emphasizing that vulnerabilities may undermine election credibility and public trust [2]. These concerns have motivated researchers to explore secure technological alternatives for modern voting systems.

2.2 Limitations of Centralized E-Voting Systems

Early online voting platforms typically rely on centralized architectures in which a single authority manages voter authentication, vote storage, and result computation. While easier to deploy, centralized systems introduce serious risks.

Research highlights that centralized infrastructures are vulnerable to cyberattacks and insider manipulation [2]. Compromise of a central database may allow unauthorized alteration or deletion of votes. Additionally, centralized systems lack transparency because voters cannot independently verify the correctness of recorded results. These weaknesses demonstrate the need for decentralized alternatives capable of improving trust and resilience.

2.3 Fundamentals of Blockchain Technology

Blockchain technology provides a decentralized ledger maintained by multiple distributed nodes [3]. Transactions are grouped into blocks and cryptographically linked, making tampering practically infeasible. This structure enhances transparency and immutability, which are essential properties for voting systems.

Consensus mechanisms ensure agreement among participating nodes regarding the validity and order of transactions [4]. Ethereum extends blockchain capabilities by supporting smart contracts; self-executing programs that automatically enforce predefined rules [5]. Modern consensus mechanisms such as Proof-of-Stake further enhance efficiency and network security [13].

These features make blockchain a promising foundation for secure and transparent voting systems.

2.4 Smart Contracts in Voting Systems

Smart contracts enable automated management of election processes without centralized control. In blockchain-based voting systems, smart contracts are used to register voters, manage candidates, record votes, and automatically compute results [6].

Research demonstrates that decentralized voting protocols enhance transparency and reduce risks of vote manipulation compared to centralized systems [7], [8]. However, vulnerabilities in smart contracts may compromise system integrity if not carefully audited [9], [16]. Since deployed contracts are difficult to modify, secure design and testing are critical.

2.5 Existing Blockchain-Based Voting Proposals

Several blockchain-based voting frameworks have been proposed. Conceptual secure voting architectures highlight the benefits of immutability and transparency [8]. Remote blockchain voting systems have also been developed to improve accessibility for geographically distant voters [10].

End-to-end verifiable election models integrate blockchain with cryptographic techniques to preserve voter privacy while maintaining public auditability [11]. Despite these advantages, studies indicate that scalability, transaction cost, and performance limitations remain key challenges in blockchain adoption [3], [13].

2.6 Role of Blockchain Wallets

Blockchain-based voting systems require secure user authentication and transaction authorization mechanisms. Digital wallets, such as MetaMask, allow users to store cryptographic keys and sign transactions securely [14].

Wallet-based authentication replaces traditional password-based systems with digital signatures, reducing risks associated with centralized credential storage. However, usability studies indicate that blockchain wallets may present challenges for non-technical users [15].

2.7 Web Interfaces in Blockchain Voting Systems

While blockchain ensures backend integrity, front-end usability significantly influences adoption. Usability engineering principles emphasize the importance of intuitive and accessible interfaces [16].

Research on blockchain-based application design highlights that poorly designed interfaces can discourage participation regardless of underlying security strength [17]. Therefore, effective web interfaces are essential for bridging complex blockchain operations with everyday users.

2.8 Security Challenges in Blockchain Voting

Although blockchain enhances transparency and tamper resistance, vulnerabilities may still exist at the application and smart contract layers. Studies on smart contract security identify potential coding flaws that attackers may exploit [9], [16].

Broader analyses of blockchain systems also reveal privacy and security risks that require comprehensive mitigation strategies [2]. Consequently, researchers recommend combining blockchain security with robust application-level protections to ensure end-to-end system integrity.

2.9 Research Gaps

Based on existing studies, several gaps remain:

1. Limited focus on practical institutional deployment.
2. Scalability and transaction cost challenges persist.
3. Usability issues affect non-technical users.
4. Balancing privacy with transparency remains complex.

These gaps highlight the need for a secure, practical, and user-friendly blockchain-based voting system suitable for institutional environments.

2.10 Conclusion

The literature demonstrates that blockchain and smart contracts significantly improve transparency, decentralization, and integrity in voting systems compared to centralized approaches. Wallet-based authentication strengthens security, while usability research emphasizes the importance of accessible design. However, challenges related to scalability, security auditing, and practical deployment remain.

The proposed project builds upon these findings to develop a practical blockchain-based voting system tailored for institutional applications.

Chapter 3- System Design

This chapter presents the architectural and operational design of the proposed blockchain-based e-voting system. The system follows a layered architecture whereas each layer performs specific responsibilities while maintaining controlled interaction with other components.

3.1 System Architecture Overview

The proposed system consists of the following major layers:

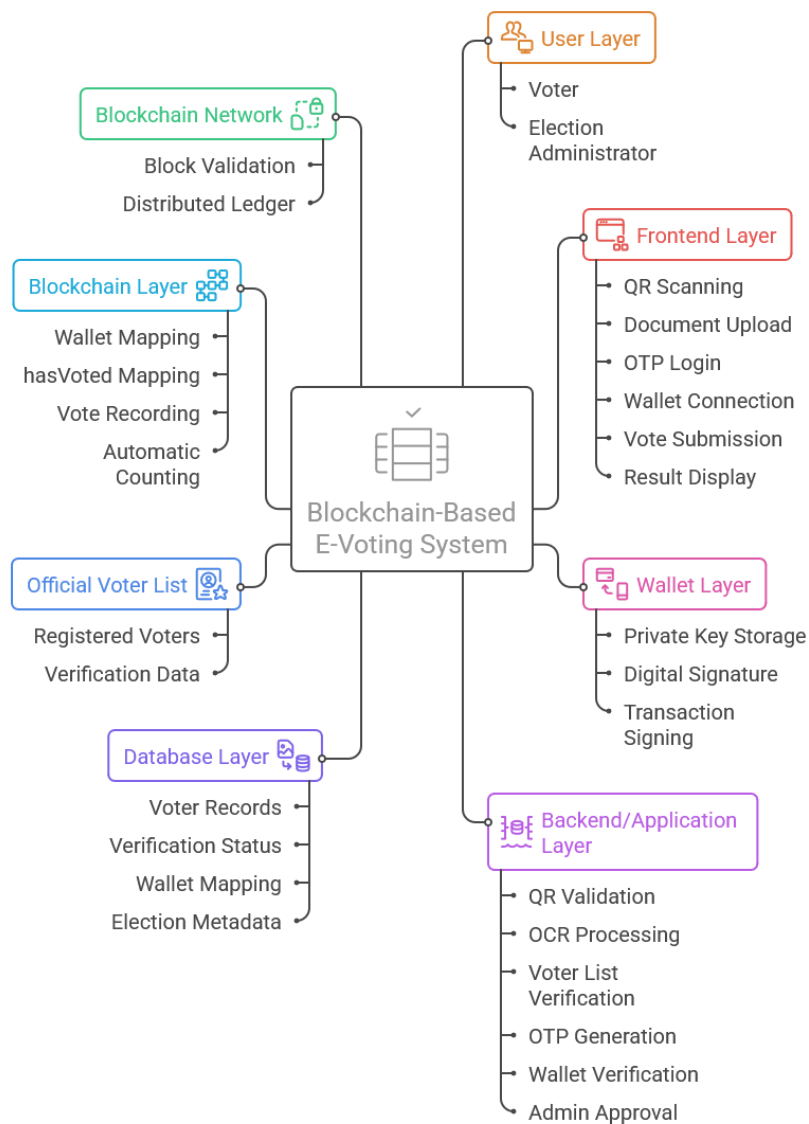


Figure 1- System Architecture

The architecture separates identity verification (off-chain) from vote storage (on-chain). Sensitive voting data is stored only on the blockchain, while auxiliary data is handled off-chain for efficiency.

3.2 Detailed System Design

3.2.1 User Layer

The User Layer includes:

- Voters: Voters register, authenticate, and cast votes
- Election Administrator: Administrators verify voter eligibility and manage elections.

3.2.2 Frontend Layer

The frontend provides the user interface for registration, login, and voting. It performs:

- QR code scanning
- Document upload
- OTP handling
- Wallet connection
- Transaction submission
- Result display

It does not store sensitive data or private keys.

3.2.3 Wallet Layer

The Wallet Layer (MetaMask) manages cryptographic identity.

Functions include:

- Secure private key storage

- Digital signature generation
- Vote transaction signing

During registration, the wallet address is bound to the verified voter. During voting, only the registered wallet is permitted to cast a vote.

3.2.4 Backend/Application Layer

The backend manages off-chain logic and verification processes:

- QR data extraction
- OCR document processing
- Validation against official voter list
- OTP generation and verification
- Wallet verification
- Admin approval workflow

It does not store or alter votes.

3.2.5 Database Layer

The database stores non-sensitive information such as:

- Registration details
- Verification status
- Wallet address mapping
- Election metadata
- System logs

Vote records are never stored in the database.

3.2.6 Blockchain Layer

The blockchain layer uses Ethereum smart contracts to:

- Register eligible wallet addresses
- Enforce one-vote-per-wallet rule
- Record votes immutably
- Automatically count votes

Before recording a vote, the contract verifies:

- Wallet is registered
- Voter has not voted
- Election is active

Votes become immutable once stored.

3.2.7 Security and Network Layer

Security is implemented through:

- HTTPS communication
- OTP authentication
- Wallet-based digital signatures
- Role-based access control
- Smart contract validation

The system uses layered authentication combining QR validation, OCR verification, official voter list matching, administrative approval, OTP authentication, and blockchain signature enforcement.

3.3 Secure Voting Workflow

The workflow is divided into two phases.

3.3.1 Registration Phase

1. Voter scans QR from organizational ID.
2. System extracts identity data.
3. User uploads ID documents for OCR verification.
4. Backend validates data against official voter list.
5. User connects MetaMask wallet.
6. Wallet address is bound to voter identity.
7. Administrator approves registration.

Upon approval, the voter becomes eligible.

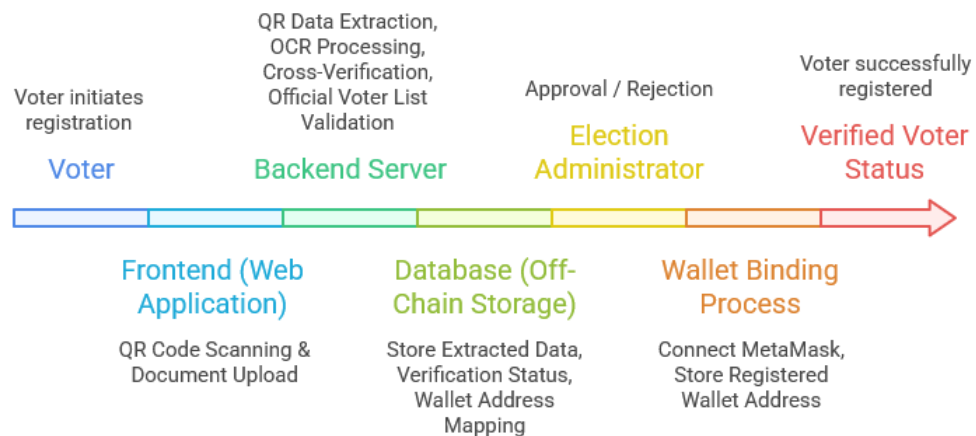


Figure 2- Registration Workflow Diagram

3.3.2 Voting Phase

1. Voter logs in using phone/email and OTP.
2. Voter connects registered wallet.

3. Wallet ownership is verified via digital signature.
4. Voter selects candidate and confirms transaction.
5. Smart contract validates eligibility and voting status.
6. Vote is recorded on blockchain.
7. Vote count updates automatically.

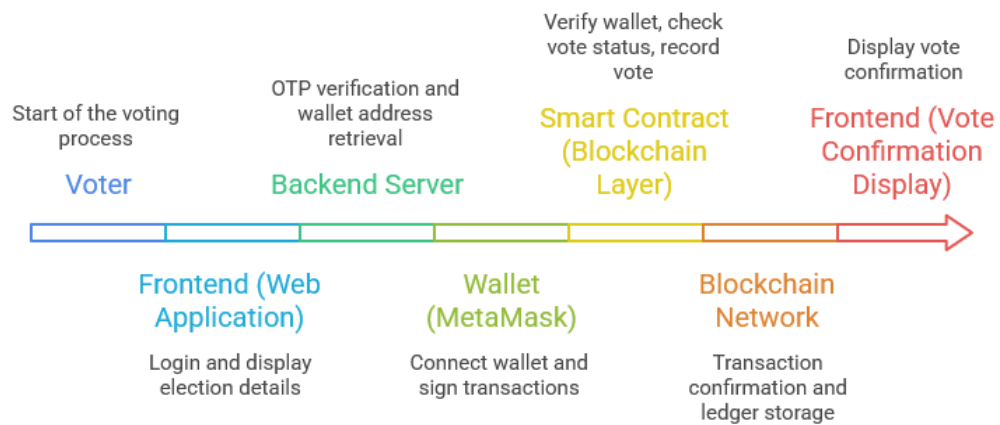


Figure 3- Voting Workflow Diagram

3.4 Data Flow Diagram (DFD)

The Data Flow Diagram illustrates how information moves within the system.

3.4.1 DFD Level 0 (Context Diagram)

The system is represented as a single process interacting with:

- Voter
- Election Administrator
- Blockchain Network

It accepts registration data, processes votes, stores transactions on blockchain, and provides results.

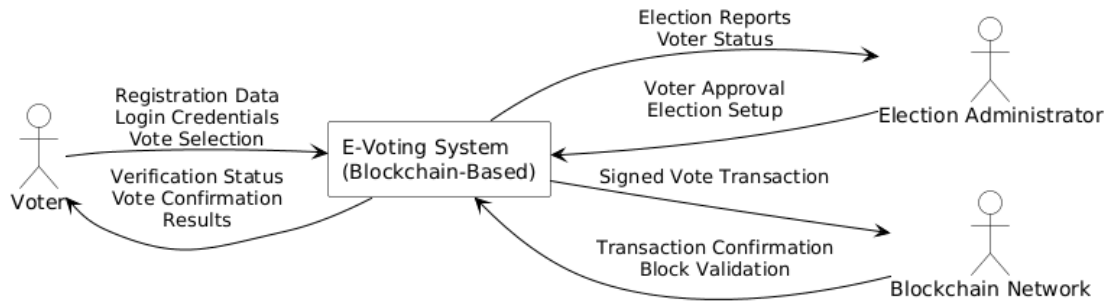


Figure 4- DFD Level 0

3.4.2 DFD Level 1 (Detailed Diagram)

This diagram expands the system into internal processes and shows detailed data movement.

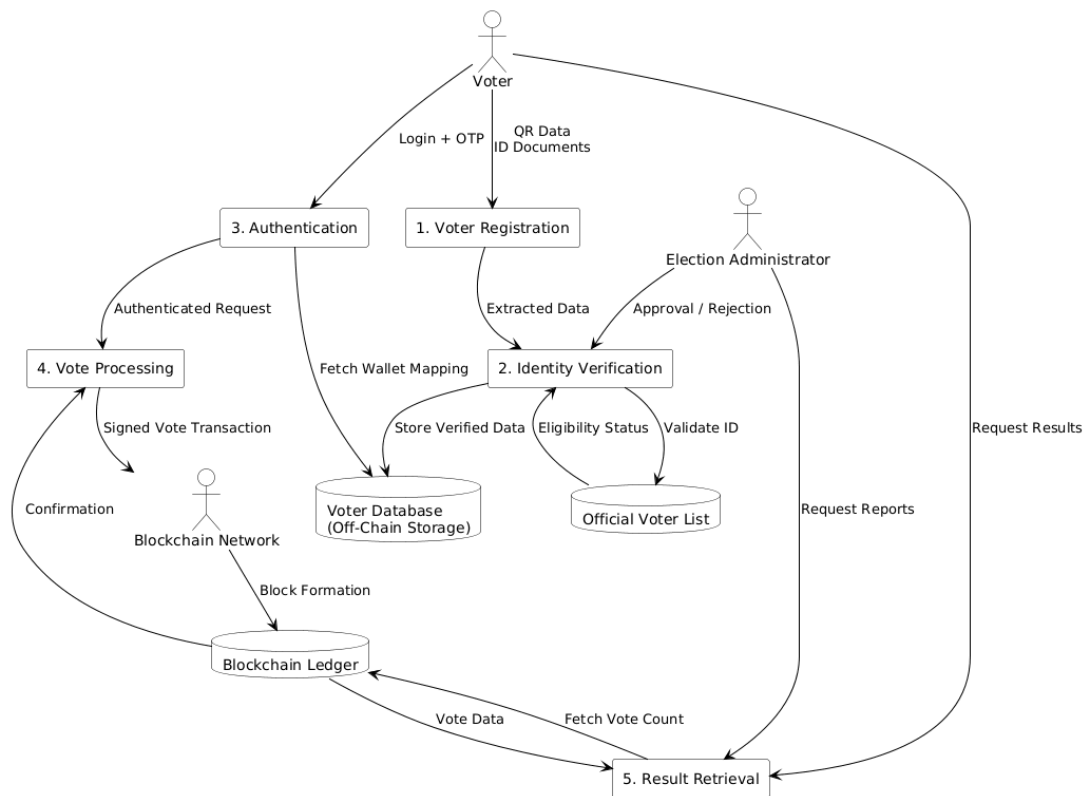


Figure 5- DFD Level 1

The DFD Level-1 diagram provides a detailed view of how data flows through the major processes of the blockchain-based online voting system.

3.5 Sequence Diagram (Workflow Representation)

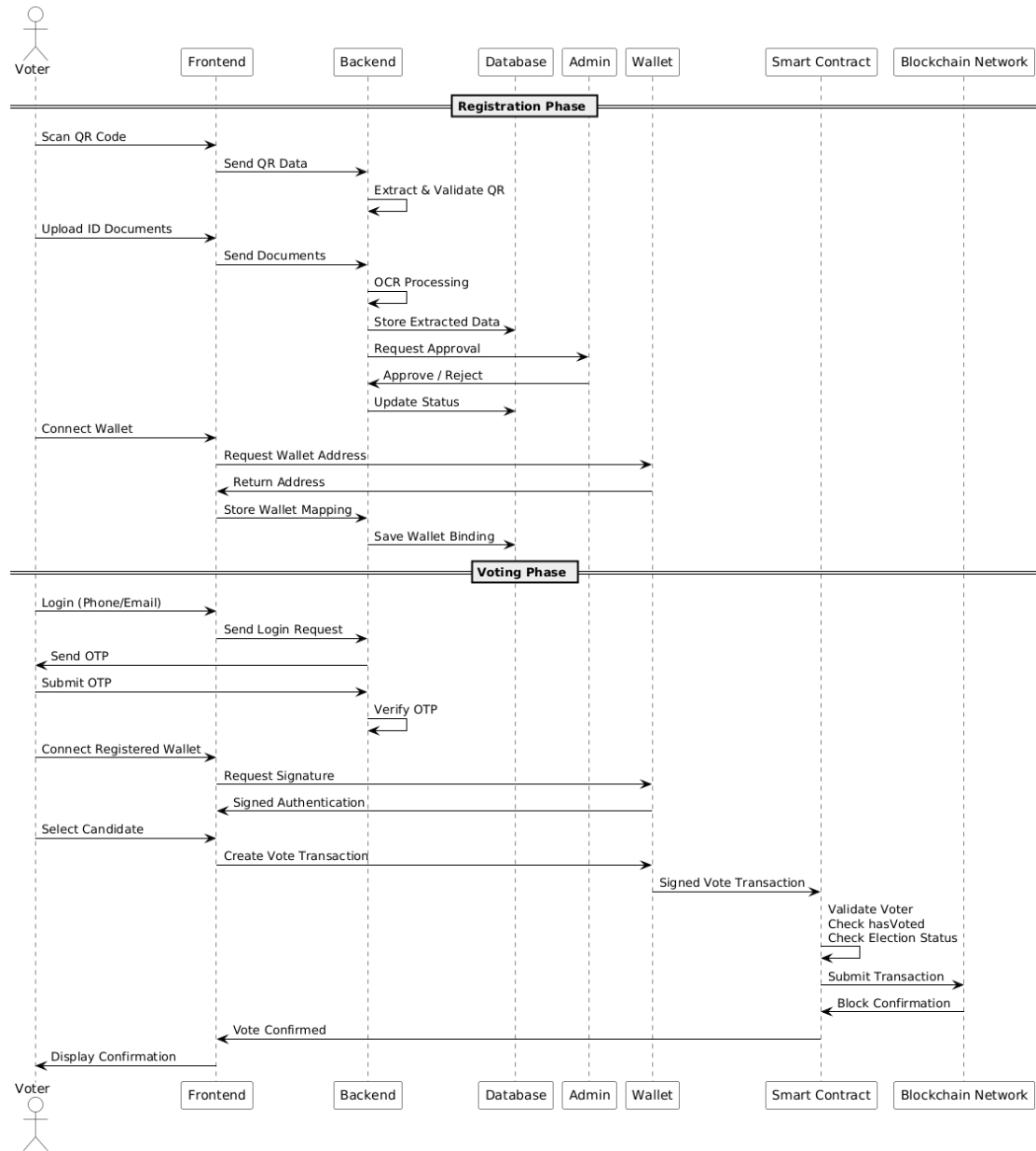


Figure 6- Complete System Workflow Sequence Diagram

3.5 Technology Stack

This section outlines the technologies used to implement the proposed e-voting system.

3.5.1 Client Layer

The client layer provides the user interface for voters and administrators.

- **React.js:** Used to build a dynamic and responsive web-based interface for registration, authentication, and voting operations.
- **HTML, CSS, JavaScript:** Used to structure and manage interactive behavior of the application.
- **Tailwind CSS / Bootstrap:** Used for responsive design and consistent user interface styling across devices.

This layer focuses on accessibility and user experience while delegating critical logic to backend and blockchain layers.

3.5.2 Backend/Application Layer

The backend manages off-chain processing, verification, and system coordination.

- **Node.js:** Implements server-side logic, API handling, and system integration.
- **Ethers.js / Web3.js:** Enables interaction with Ethereum smart contracts for transaction submission and data retrieval.
- **Tesseract OCR (or similar):** Extracts identity information from uploaded voter documents.

The backend performs QR validation, OCR verification, OTP authentication, wallet mapping, and administrative approval workflows. It does not store or modify votes.

3.5.3 Blockchain Layer

The blockchain layer ensures decentralization, transparency, and immutability.

- **Ethereum Blockchain:** Records vote transactions in a distributed ledger.
- **Solidity:** Used to develop smart contracts for voter registration, vote validation, and automatic counting.
- **Ganache:** Used for local blockchain simulation and smart contract testing during development.

This layer enforces voting rules such as one-vote-per-wallet and ensures tamper-resistant vote storage.

3.5.4 Database Layer

The database stores non-sensitive and auxiliary system data.

- **Firebase / MySQL:** Stores voter registration details, verification status, wallet mappings, election metadata, and system logs.

No vote data is stored in the database to preserve blockchain integrity.

3.5.5 Security Layer

Security mechanisms are integrated across system layers.

- **MetaMask Wallet Authentication:** Provides cryptographic identity and digital signature verification.
- **National ID & Organizational ID Verification:** Ensures voter legitimacy during registration.
- **OTP-Based Authentication:** Adds multi-factor login security.

Together, these mechanisms provide layered security against impersonation, unauthorized access, and multiple voting attempts.

Chapter 4- Expected Output

The proposed blockchain-based e-voting system is expected to deliver a secure, transparent, and efficient platform for conducting institutional elections. By integrating blockchain technology with structured identity verification mechanisms, the system addresses key limitations of centralized and traditional voting approaches.

The system will provide a functional web-based interface that enables verified voters to participate remotely. Voter legitimacy is ensured through QR-based identity extraction, OCR document verification, official voter list validation, administrative approval, OTP authentication, and wallet binding. This layered approach guarantees that only eligible and authorized users can cast votes.

Votes are recorded as digitally signed transactions on the Ethereum blockchain through smart contracts. Once confirmed, each vote becomes immutable and publicly verifiable. Smart contracts enforce voting rules such as one-vote-per-wallet and active election validation, preventing duplication and unauthorized participation.

The system also supports automatic vote counting. Since votes are processed directly on-chain, results can be retrieved instantly without manual tallying. This improves accuracy, reduces human error, and enhances trust in the election process.

Administrative functionality includes voter verification management, election setup, candidate registration, and monitoring of voting activity. Automation of verification and vote processing reduces operational workload and increases efficiency.

Non-sensitive data such as voter records and election metadata are stored in an off-chain database to improve performance, while all vote transactions remain securely stored on the blockchain.

Overall, the project delivers a complete proof-of-concept implementation of a secure blockchain-based voting system suitable for institutional environments.

Chapter 5- Conclusion

This project presents the design and implementation of a secure and transparent blockchain-based e-voting system suitable for institutional and organizational elections. By integrating web technologies, QR and OCR-based identity verification, OTP authentication, wallet-based cryptographic signatures, and Ethereum smart contracts, the system ensures that votes are securely authorized, recorded, and counted.

The separation between off-chain identity verification and on-chain vote storage enhances both efficiency and security. Smart contracts enforce critical rules such as one-vote-per-wallet and election validity, while blockchain immutability guarantees that recorded votes cannot be altered.

The system improves transparency through publicly verifiable results and enhances accessibility by enabling remote participation via a web interface. At the same time, layered security mechanisms significantly reduce the risks of impersonation and unauthorized voting.

In conclusion, the proposed system demonstrates the feasibility and effectiveness of applying blockchain technology to modernize institutional voting processes. It provides a secure, transparent, and reliable framework that can serve as a foundation for future research and practical deployment in controlled election environments.

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